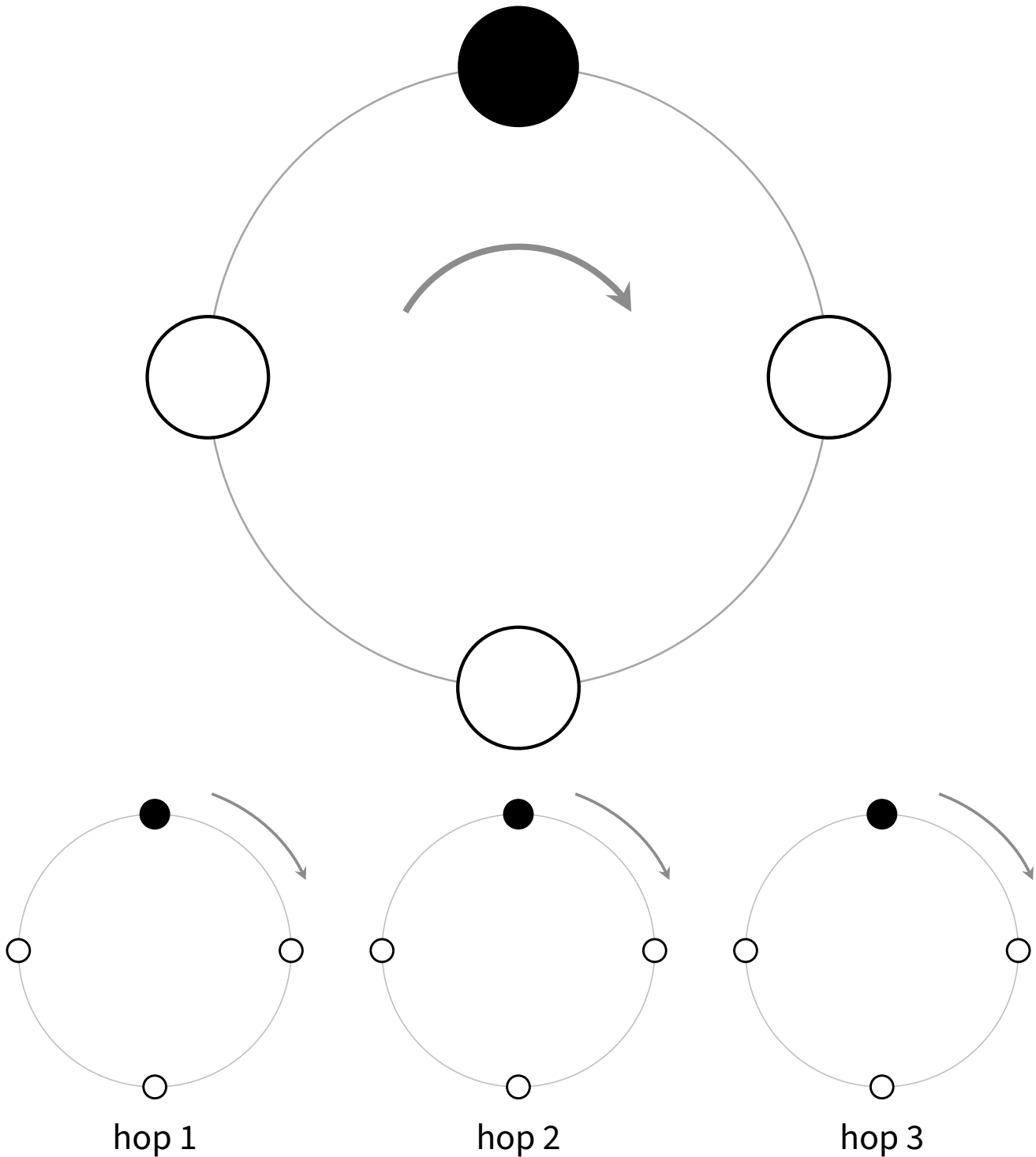
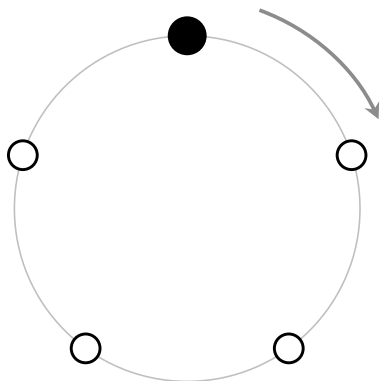
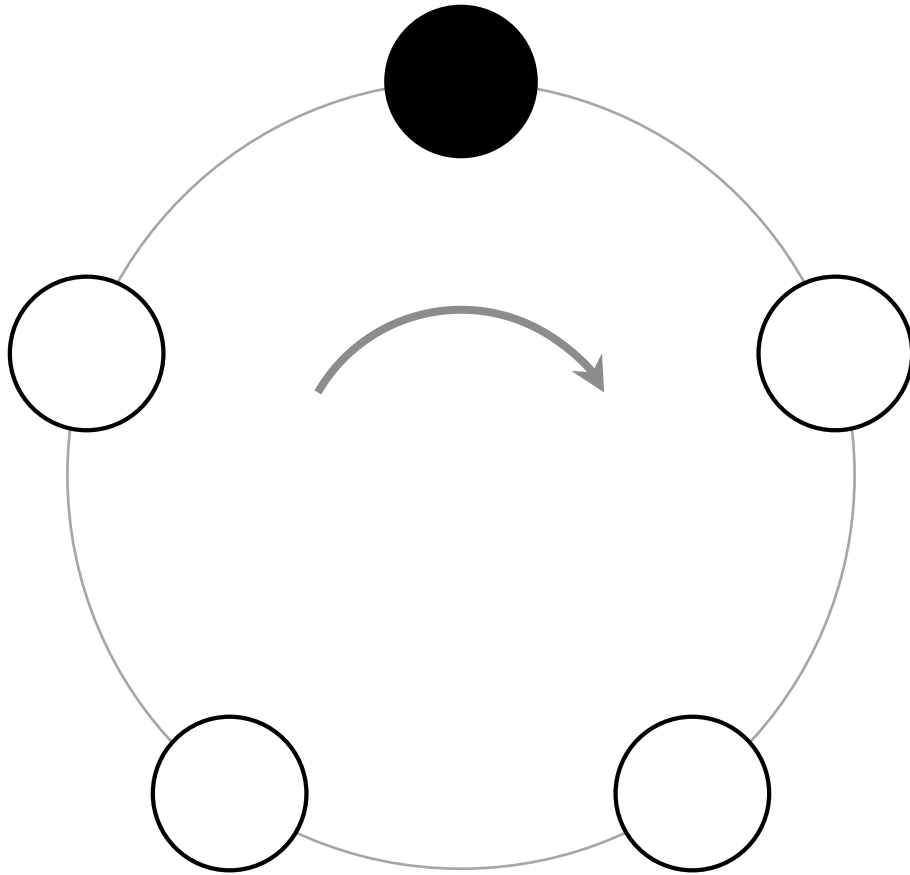


To hop 2, move the counter 2 dots along the way the arrow points.

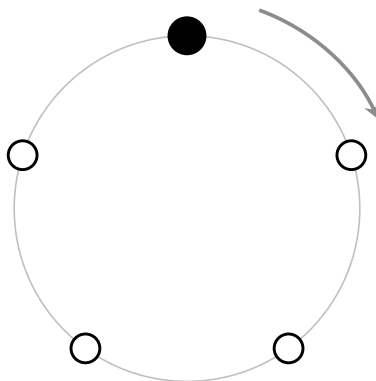
Problem 1: Put a counter on the black dot and hop it until it is back. Draw each hop on a small ring, and circle each hop that lands on every dot.



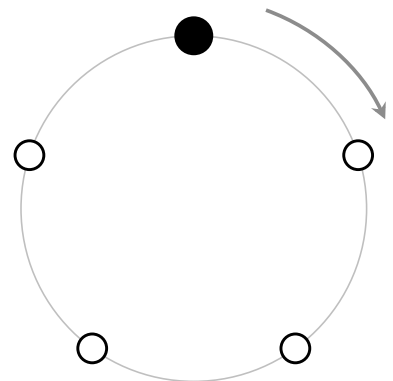
Problem 2: Do the same with this ring of 5 dots.



hop 1

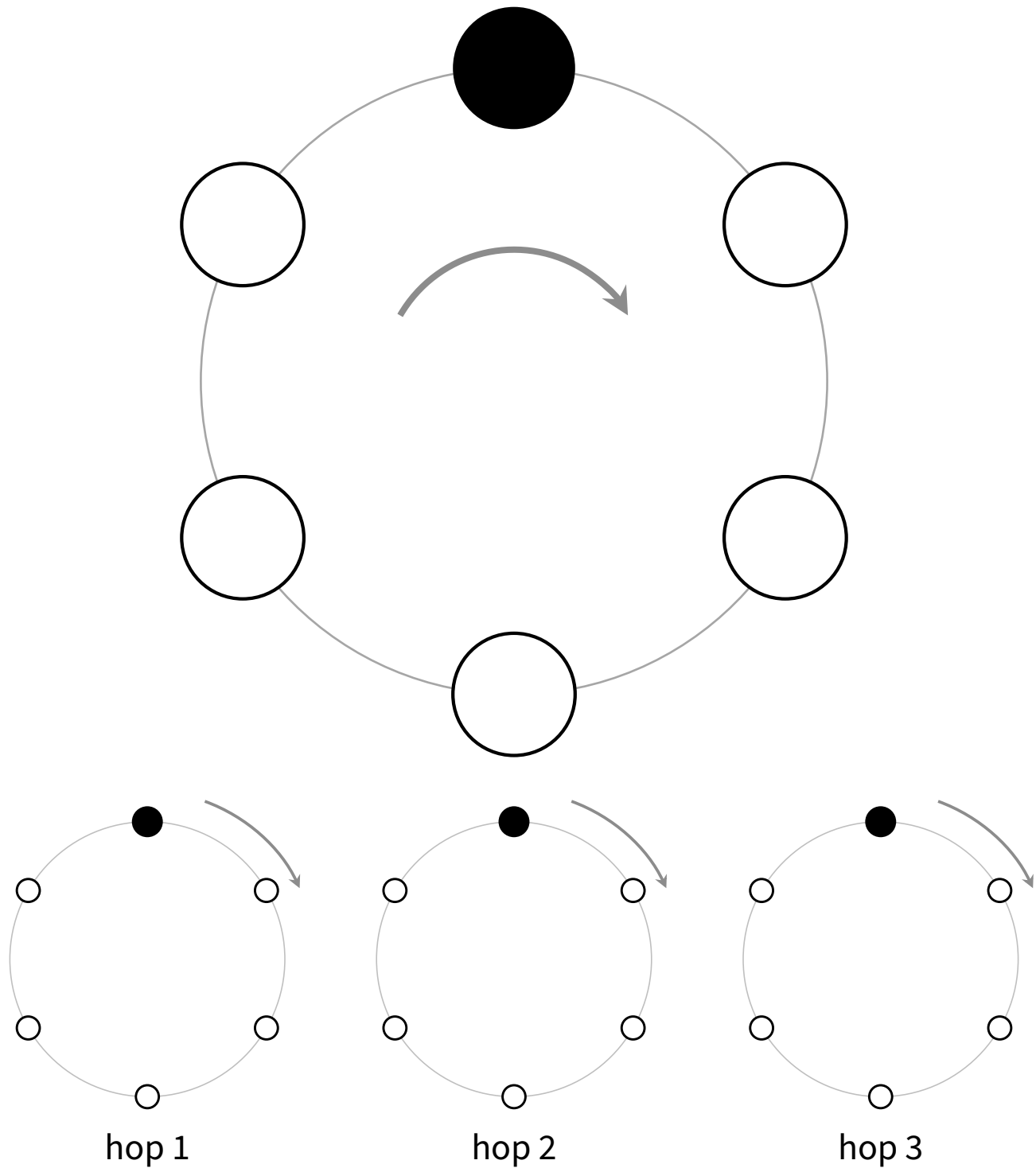


hop 2

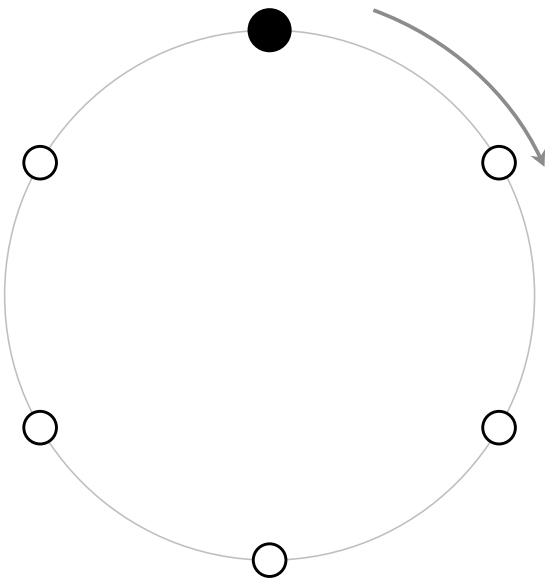


hop 3

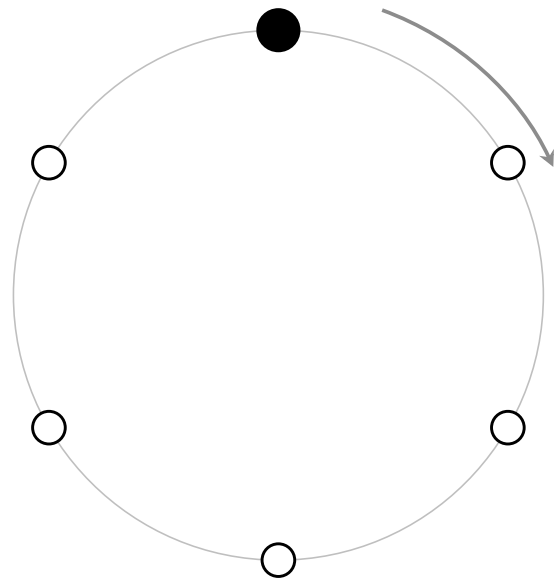
Problem 3: Do the same with this ring of 6 dots.



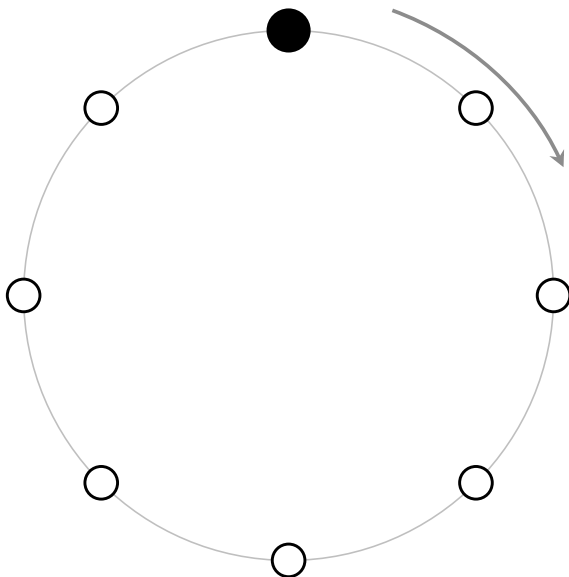
Problem 4: Draw the hops from the black dot until you are back. If a dot has no line, start again there with the same hop. Write in the box how many times you started.



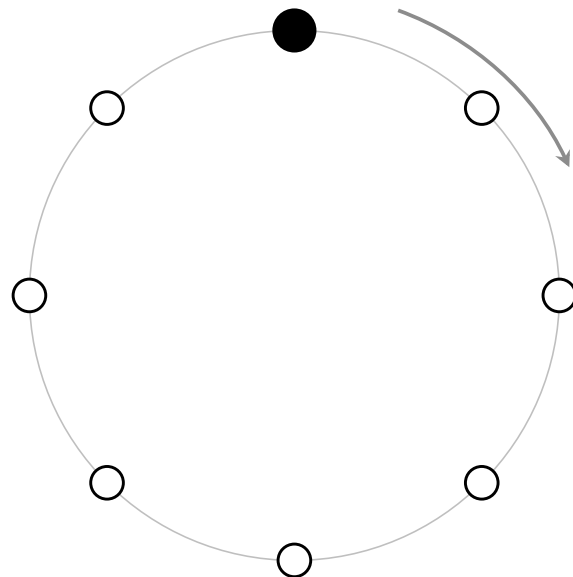
hop 2



hop 3

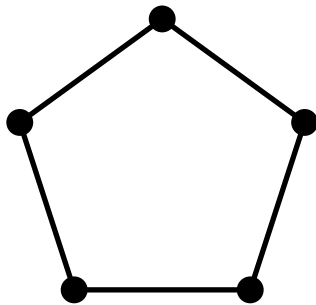


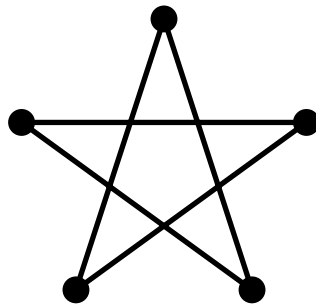
hop 2

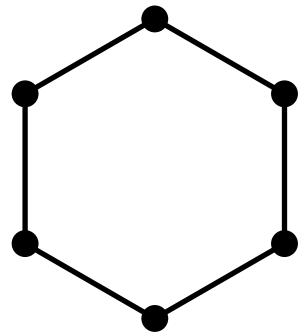


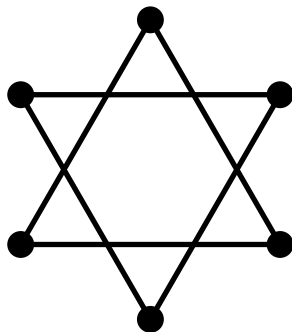
hop 3

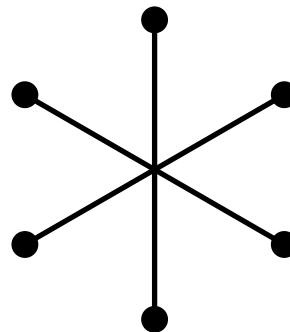
Problem 5: Find every hop from 1 to 5 that draws each picture. Write the hops under the picture.

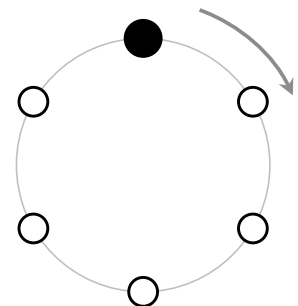
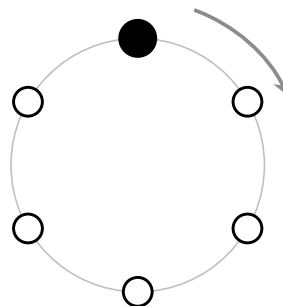
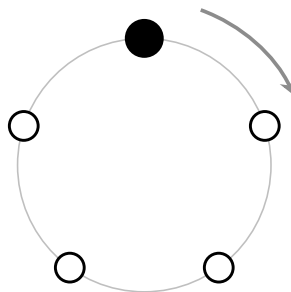
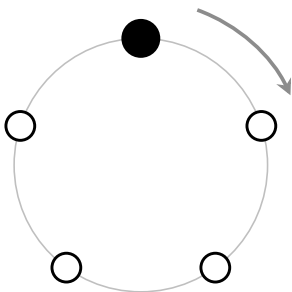




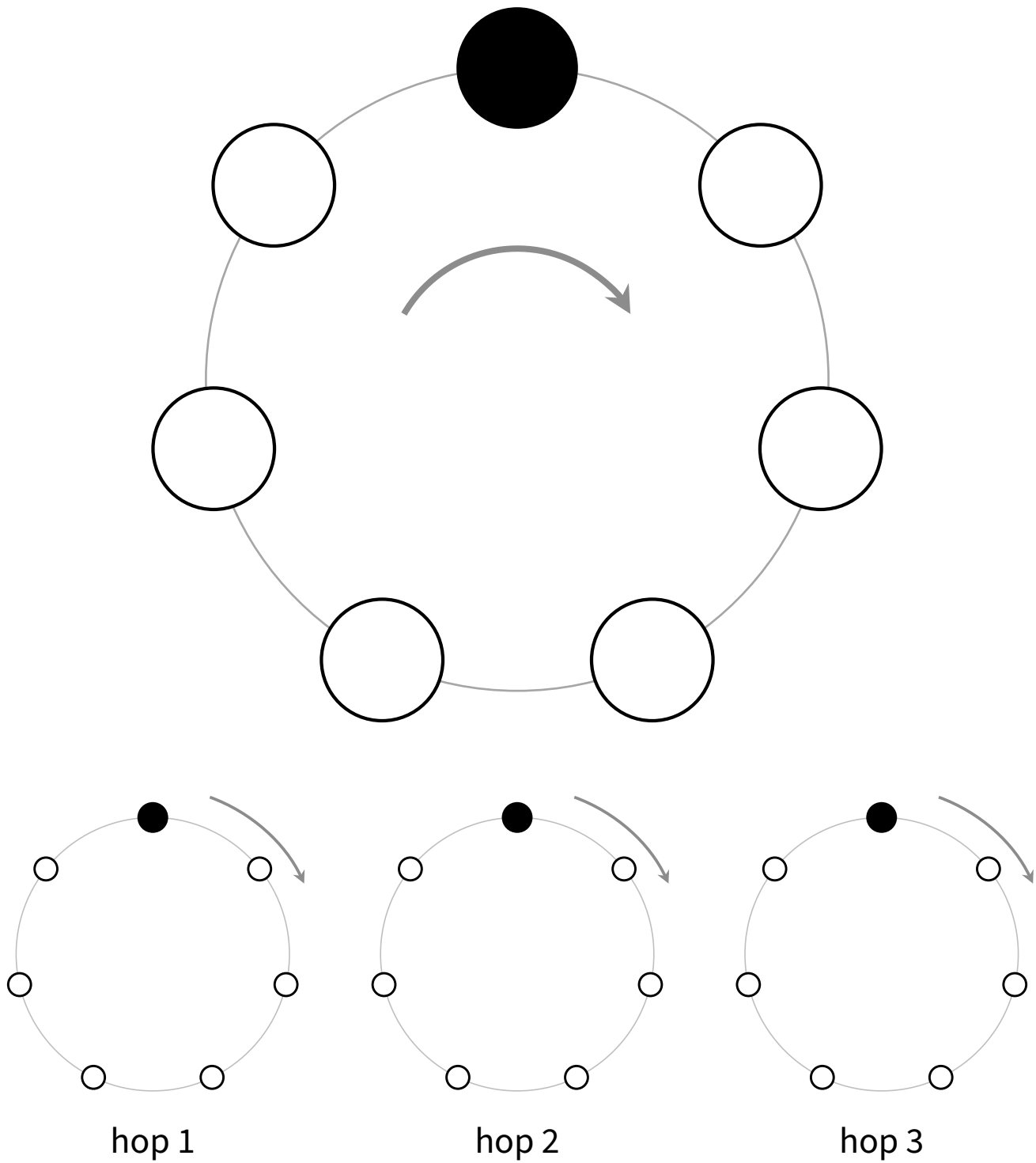




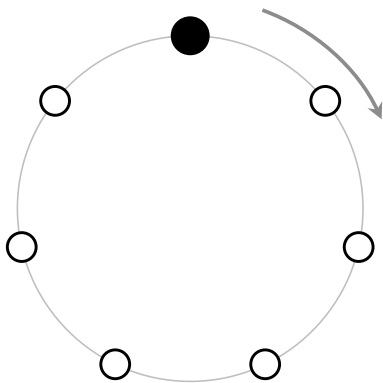
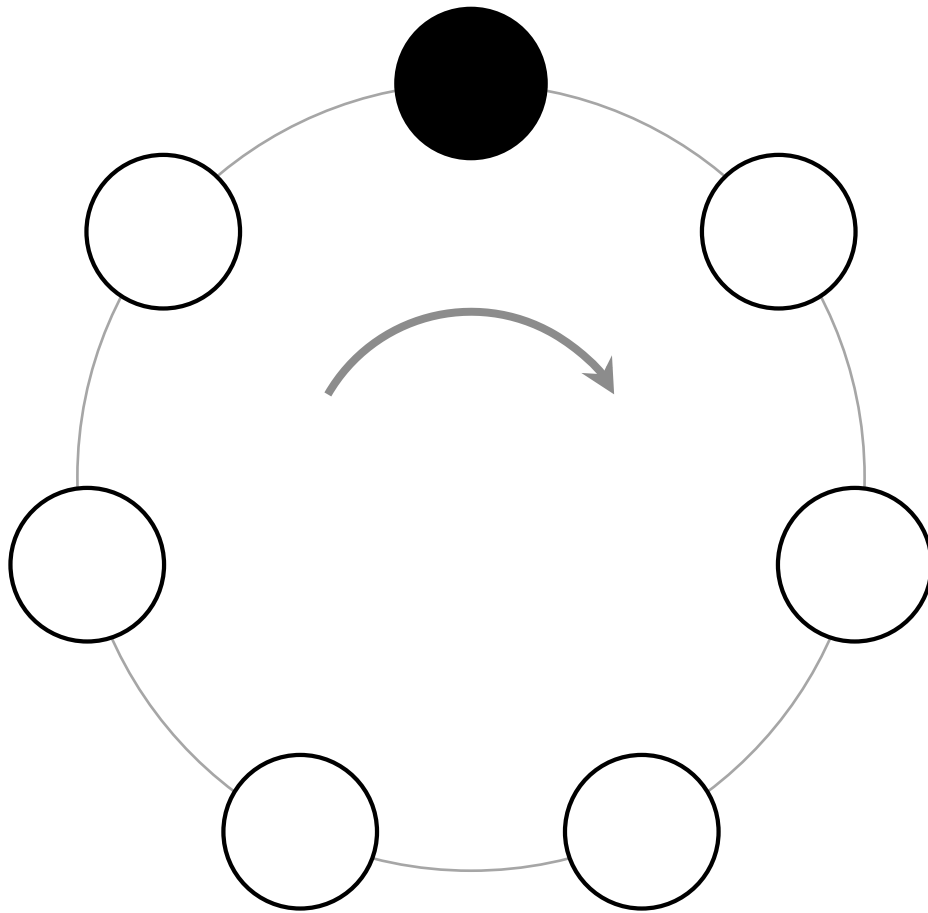




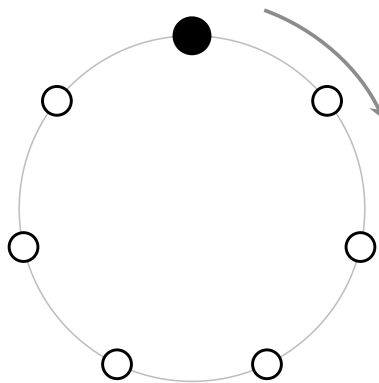
Problem 6: Put a counter on the black dot and hop it until it is back. Draw each hop on a small ring, and circle each hop that lands on every dot.



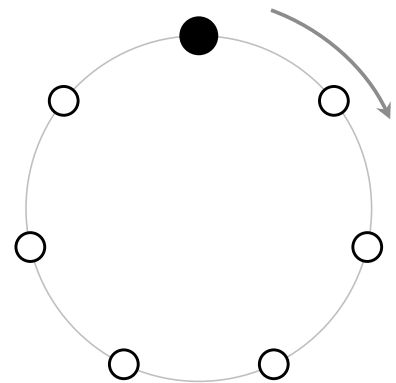
Problem 7: Do the same with hop 4, hop 5 and hop 6 on this ring of 7 dots.



hop 4

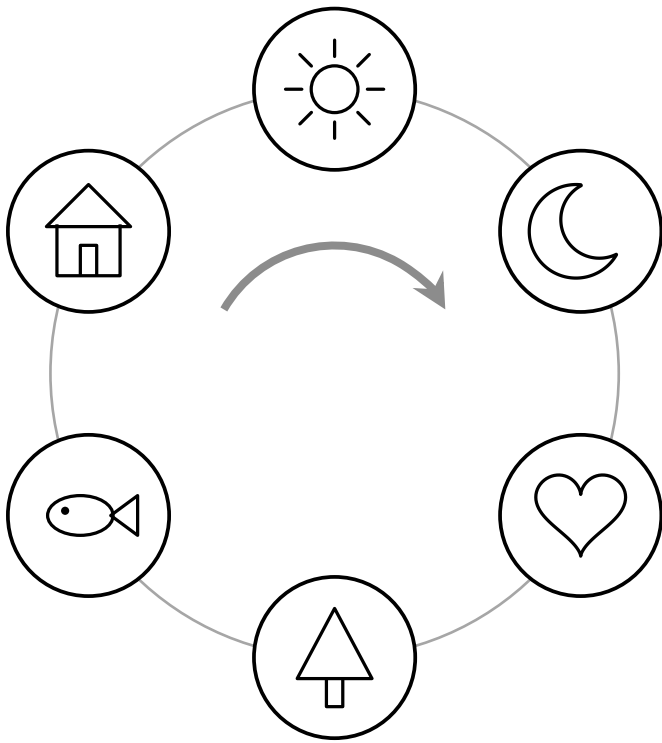


hop 5

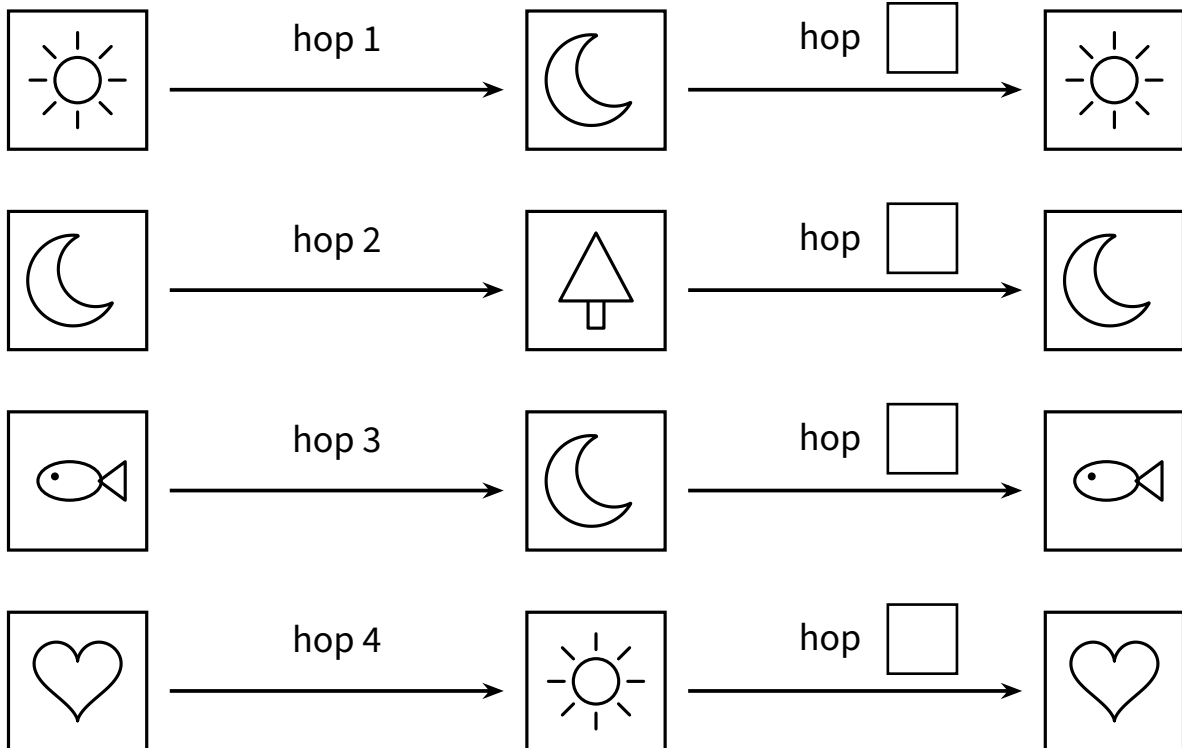
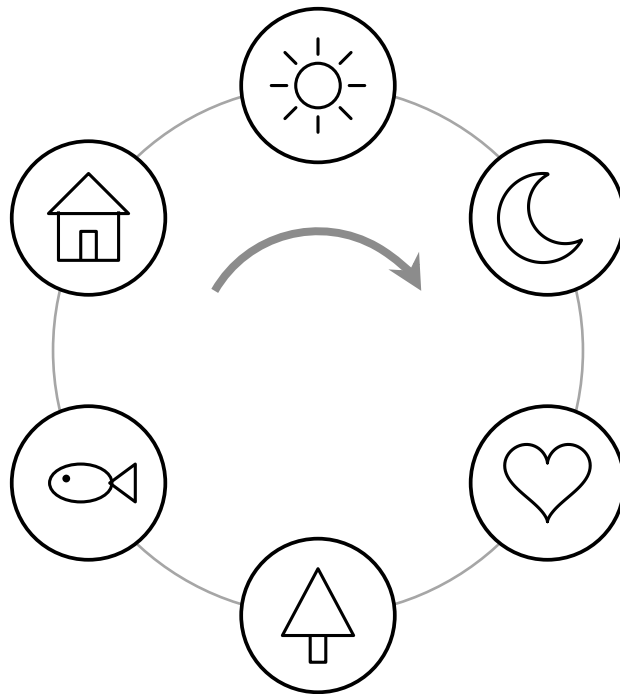


hop 6

Problem 8: Hop 2 changes each picture into the picture 2 places along the arrow. Use hop 2 on each row to make the next row, until you get the top row again.



Problem 9: Which hop changes the picture back to what it was? Write that hop in the box.



Problem 10: A secret hop changed the sun into the tree. Draw what the same hop changes each of these pictures into.

